

Business Context: Problem Definition

Document: 01-business-context/problem-definition.md | Project: MsingiPack Academy CRO & Analytics Framework | Date: August 2026

Executive Summary

MsingiPack Academy faces severe performance tracking disconnects and conversion bottlenecks across its digital marketing funnel. While paid acquisition campaigns (Meta Ads) drive high top-of-funnel reach (130,088 impressions) and ad clicks (7,486 clicks), operational infrastructure defects result in massive lead drop-offs, distorted analytics, and uncredited revenue.

Key underlying friction points include:

- **Moodle Security Redirects:** Mobile ad traffic intended for course previews (preview_courses.php) is systematically redirected to the generic homepage (/), stripping UTM parameters and driving an 11-second average session duration.
- **Superficial Tag Triggers:** Meta Ads conversion tags trigger on button click rather than verified database entry, masking a 53.9% drop-off between intent and account creation.
- **Post-Registration Activation Cliff:** 92.2% of registered users (3,183 out of 3,452) never log in after profile creation.
- **Flawed Attribution & Revenue Tracking:** GA4 underreports actual financial revenue (KES 9,900 logged vs. KES 19,333 actual M-PESA/Bank ledger) while duplicating purchase events for returning users (averaging 8.25 events per buyer).

Current State vs. Ideal State

Dimension	Current State	Ideal State	Gap / Business Impact
Landing Page Routing	87.2% of traffic lands on homepage (/) due to Moodle guest security redirects.	Ad traffic lands directly on preview_courses.php with preserved UTM context.	Confused users exit in 11 seconds; 74.5% of sessions mislabeled as 'Paid Other'.
Conversion Accuracy	Meta fires registration conversions on CTA button clicks (7,486 reported).	Conversions fire strictly on confirmed Moodle database record creation (3,452 actual).	53.9% phantom conversion reporting (4,034 unconfirmed leads billed as conversions).
Lead Activation	92.2% idle rate (3,183 registered accounts never log in).	Automated post-signup activation sequence drives immediate portal entry.	High-intent parents sit dormant in the database without completing enrollment.
Attribution Persistence	Click IDs (fbclid) and UTM parameters stripped upon landing/redirect.	Persistent 30-day cross-subdomain cookie tracking preserves attribution.	Delayed sales attributed to generic 'Direct' or 'Organic' instead of Meta Ads.
Revenue Tracking	GA4 logs KES 9,900.00; duplicate pixels fire 8.25x per buyer on refresh.	Transaction-deduplicated pixel fires logging exact M-PESA/Bank revenue (KES 19,333.00).	Inability to evaluate true Return on Ad Spend (ROAS) or campaign profitability.

Revenue Leakage Hypotheses

H1: Top-of-Funnel Landing Page Trap (Security Redirect)

Mechanic: Ad campaigns direct parents to inspect course content at preview_courses.php. Moodle's default security configuration forces unauthenticated mobile visitors onto the root homepage (/).

Impact: 4,084 total sessions forced to homepage with an 11-second average engagement time. Tracking parameters are stripped in the redirect chain, categorizing 4,179 sessions (74.5%) under a generic 'Paid Other' bucket in GA4.

H2: Mid-Funnel Sign-up Form Drop-Off & Tag Misalignment

Mechanic: The Meta Ads pixel fires a registration conversion when a user presses 'Register' or 'Get Started'.

Impact: 4,034 prospective parents click the CTA but fail to complete multi-step profile creation, creating a 53.9% discrepancy between reported conversions (7,486) and actual database accounts (3,452).

H3: Post-Registration Onboarding Bottleneck

Mechanic: Parents complete form registration but encounter friction (missing immediate credentials, lack of SMS/WhatsApp follow-up, or unclear next steps) before entering the learning environment.

Impact: 3,183 out of 3,452 registered accounts (92.2%) remain idle and have never logged in, leaving active users at a critical low of 269 (7.8%).

H4: Bottom-of-Funnel Attribution Decay & Duplicate Sales Events

Mechanic: Parent purchase cycles are delayed; parents inspect courses, leave, and return days later via Direct or Search routes.

GA4 fails to maintain fbclid parameters across subdomains. Simultaneously, receipt page refreshes re-trigger the GA4 purchase tag.

Impact: GA4 overcounts individual buyer transactions (33 purchase events across 4 buyers) while missing historical sales entirely, causing a KES 9,433.00 deficit against actual M-PESA payment receipts.

Scope & Boundaries

In-Scope:

- Re-architecting Google Tag Manager (GTM) data layer triggers for Moodle LMS.
- Resolving Moodle mobile security redirect rules to allow direct access to preview_courses.php.
- Aligning Meta Ads and GA4 conversion triggers with verified backend database creation.
- Implementing persistent first-party cookie/session attribution (fbclid, UTM tracking across subdomains).
- Setting up transaction ID deduplication on receipt pages for M-PESA and Bank payment flows.
- Granular form step tracking to isolate sign-up form abandonment points.

Out-of-Scope:

- Changes to Moodle core LMS PHP codebase beyond analytics hook integrations.
- Direct modifications to course video content or curriculum structure.
- Offline/manual sales activities not routed through M-PESA or Bank payment gateways.